Objectives

1. For a bond, compute the
   - price and
   - (implied) yield
2. For a stock, compute the
   - price,
   - (implied) expected dividend growth rate

Treasuries

A T-year Treasury makes semi-annual payments.

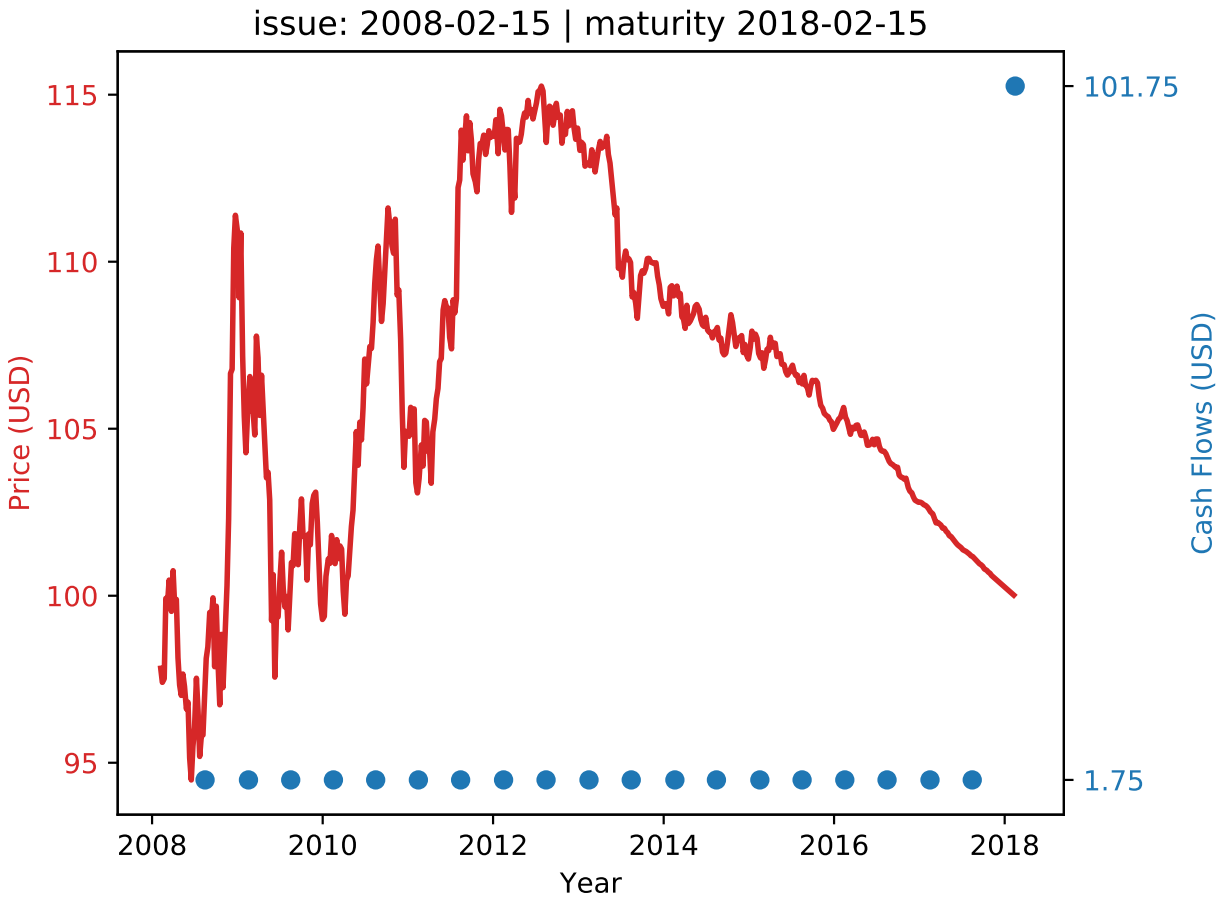
0	1	2	$\cdots$	$2T$
$-P$	$cF/2$	$cF/2$	$\cdots$	$cF/2 + F$

where $c \geq 0$ is the coupon rate (APR)

$$P \overset{\text{should}}{=} \frac{cF/2}{(1+y/2)} + \frac{cF/2}{(1+y/2)^2} + \frac{cF/2}{(1+y/2)^3} + \cdots + \frac{cF/2}{(1+y/2)^{2T}} + \frac{F}{(1+y/2)^{2T}}$$
$$= \frac{cF/2}{y/2}\left(1 - \frac{1}{(1+y/2)^{2T}}\right) + \frac{F}{(1+y/2)^{2T}}$$

where y is the yield (APR).

From the Data



Treasury Checklist

- ✓ Yields and coupon rates are quoted as APRs, so divide them by the number of payments per year to get the per-period rate
- ✓ Multiply the number of years by the number of payments per year to get the number of periods
- ✓ The future value is the face value
- ✓ Remember: if the coupon rate is greater than (less than) the yield, the price is greater than (less than) the face value
- ✓ **Use four decimal places!**

Example

On 2012-01-31, the yield on a 100 USD 5.0-year Treasury that matured on 2015-01-31 was 0.3035%. It paid a 2.25% coupon. What should it have traded for? (Hint: the number of payments up to and including maturity is 6.0)

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$$
\begin{aligned}
P_0 \overset{\text{should}}{=} & -\mathrm{PV}(\\
& \quad \mathrm{N} = 6.0, \\
& \quad \mathrm{I/Y} = 0.3035\%/2, \\
& \quad \mathrm{PMT} = (2.25\% \times 100)/2, \\
& \quad \mathrm{FV} = 100) \\
= & \ 105.8086 \text{ USD}
\end{aligned}
$$

Example

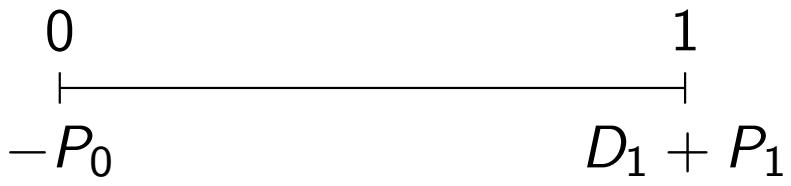
On 2012-07-31, a 5.0-year Treasury that matured on 2013-07-31 traded for 103.1758 USD. It paid a 3.375% coupon. What was the yield? (Hint: the number of payments up to and including maturity is 2.0)

Example

On 2012-07-31, a 5.0-year Treasury that matured on 2013-07-31 traded for 103.1758 USD. It paid a 3.375% coupon. What was the yield? (Hint: the number of payments up to and including maturity is 2.0)

$$
\begin{aligned}
y_0 &\overset{\text{should}}{=} \text{RATE}(\\
&\qquad \text{N} = 2.0,\\
&\qquad \text{PV} = -103.1758,\\
&\qquad \text{PMT} = (3.375\% \times 100)/2,\\
&\qquad \text{FV} = 100)\\
&= 0.1946\%
\end{aligned}
$$

Stocks



- P_1 is the expected price in one period (usually one quarter)
- r is the required total rate of return (usually quarterly)
- D_1 is the expected dividend
- The price <u>should</u> be

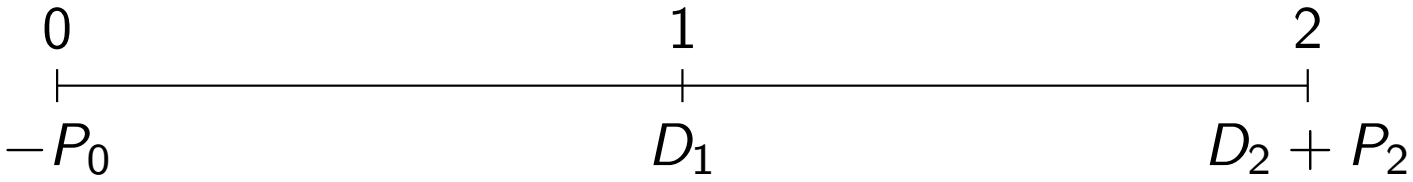
$$P_0 \overset{\text{should}}{=} \frac{D_1 + P_1}{1 + r}.$$

Didend Yields, Capital Gains, and Total Returns

$$r \overset{\text{should}}{=} \frac{D_1 + P_1}{P_0} - 1 = \underbrace{\frac{D_1}{P_0}}_{\text{Didend Yield}} + \underbrace{\frac{P_1 - P_0}{P_0}}_{\text{Capital Gain Rate}}$$

- The <u>dividend yield</u> is the expected annual dividend of the stock divided by its current price
- The <u>capital gain rate</u> is the percent by which the stock price increases from the current period to the next
- The <u>total return</u> is the sum of the dividend yield and the capital gain rate

Stocks



- The prices should be

$$P_0 \overset{\text{should}}{=} \frac{D_1}{1+r} + \frac{D_2 + P_2}{(1+r)^2} \text{ and } P_1 \overset{\text{should}}{=} \frac{D_2 + P_2}{1+r}.$$

- Does the two-year investor value the stock differently than the one-ear investor? **No!**

$$P_0 = \frac{D_1}{1+r} + \frac{D_2 + P_2}{(1+r)^2} = \frac{D_1}{1+r} + \frac{P_1}{1+r} = \frac{D_1 + P_1}{1+r}$$

The Dividend Discount Model

- It doesn't matter whether the investor wants to hold the stock for one quarter, two quarters, or infinitely many quarters: she is always willing to pay the same price P_0
- In particular, the investor who holds the stock for infinitely many periods is willing to pay

$$P_0 \overset{\text{should}}{=} \frac{D_1}{1+r} + \frac{D_2}{(1+r)^2} + \frac{D_3}{(1+r)^3} + \cdots$$

  for the stock
- Implication: we don't need to know the prices P_1, P_2, ..., just the dividends D_1, D_2, D_3, ...

Constant Dividend Growth

Simple model: dividends grow at a constant rate g

0	1	2	3	$\cdots$
$-P_0$	$(1+g)D_0$	$(1+g)^2 D_0$	$(1+g)^3 D_0$	$\cdots$

where D_0 is the dividend that was <u>just</u> paid

$$P_0 \overset{\text{should}}{=} \frac{(1+g)D_0}{(1+r)} + \frac{(1+g)^2 D_0}{(1+r)^2} + \frac{(1+g)^3 D_0}{(1+r)^3} + \cdots = \frac{(1+g)D_0}{r-g}$$

Constant Dividend Growth

- The Dividend Discount Model says that

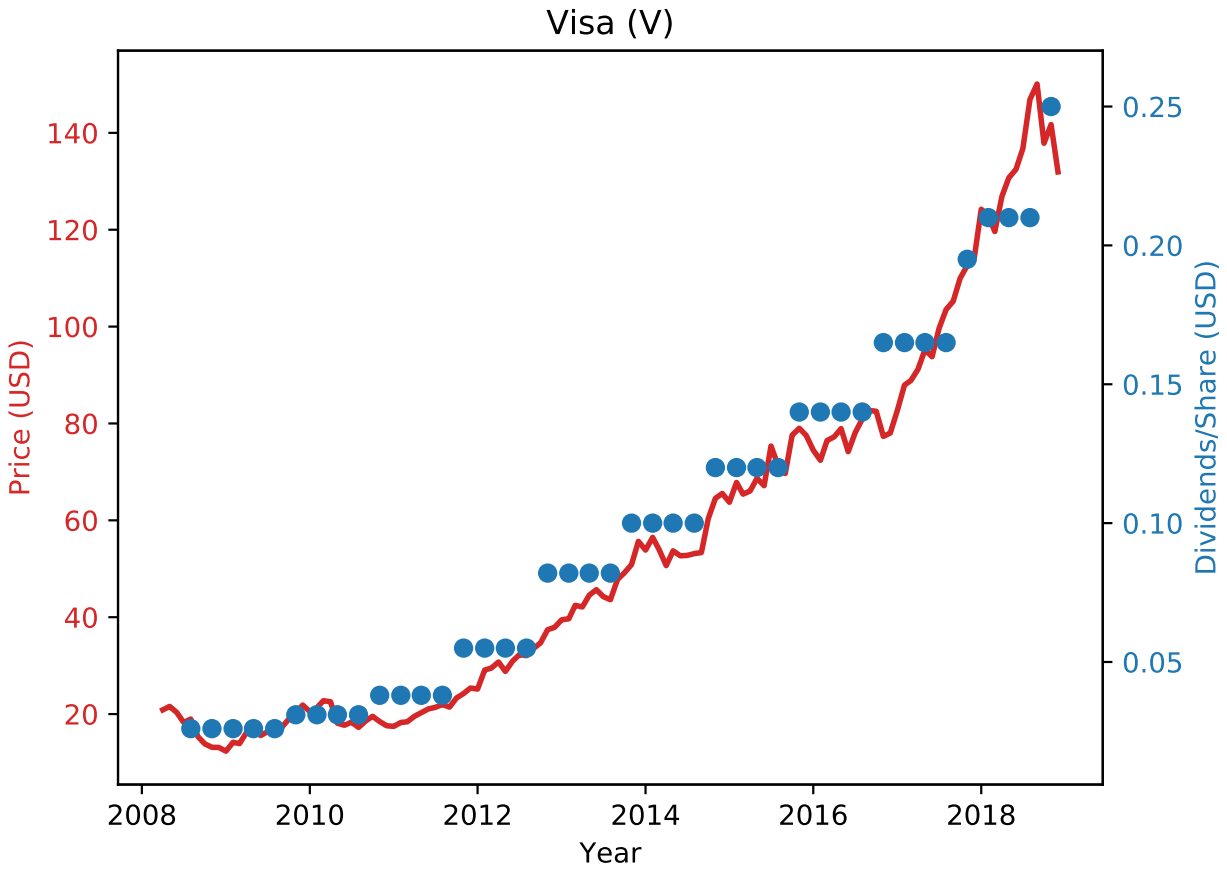
$$P_0 \overset{\text{should}}{=} \frac{(1+g)D_0}{r-g} = \frac{D_1}{r-g}$$

- Awesome interpretation!

$$\overbrace{\frac{D_1}{P_0} + g}^{\text{Specific Formula}} \overset{\text{should}}{=} r \overset{\text{should}}{=} \overbrace{\underbrace{\frac{D_1}{P_0}}_{\text{Didend Yield}} + \underbrace{\frac{P_1 - P_0}{P_0}}_{\text{Capital Gain Rate}}}^{\text{General Formula}}$$

- In this case, the <u>growth rate</u> g is the <u>capital gain rate</u>

From the Data



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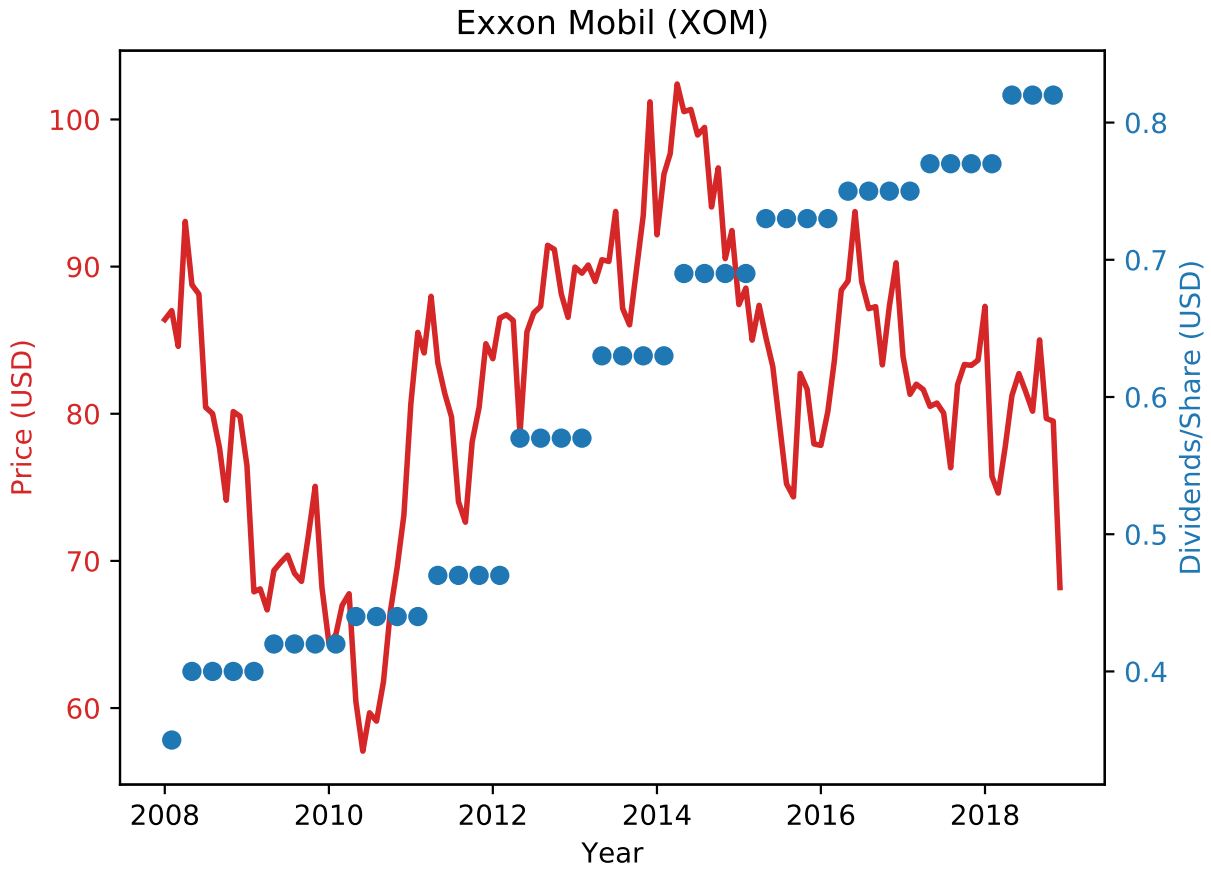

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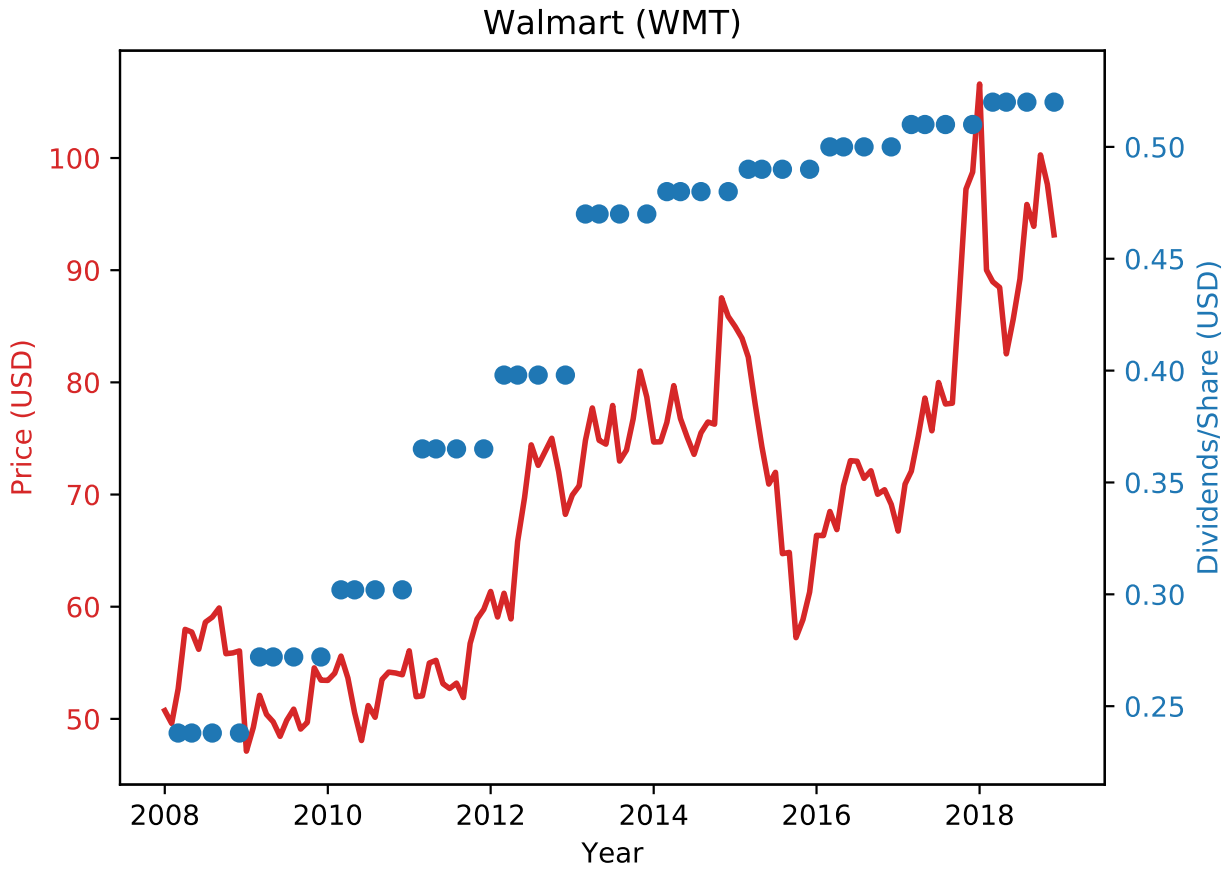

From the Data

Does the constant dividend growth assumption comport with reality?

- Verizon's dividends are (generally) constant across the four quarters in each year, but increase from the fourth quarter of one year to the first quarter of the next
- Exxon's dividends follow a similar pattern to those of Verizon, <u>despite</u> the much greater volatility in its stock price
- Walmart's dividends follow a similar pattern to those of Verizon and Exxon, <u>but</u> the annual growth in the dividend rate seems to kink in 2013

There are some issues...

Stock Checklist

- ✓ The required rate of total return should have the same horizon as that of the dividends, so if returns are quoted in months, say r_m, compute the quarterly return as follows:

$$r_q = (1 + r_m)^3 - 1 \tag{1}$$

- ✓ The growth rate should also have the same horizon as that of the dividends
- ✓ **Use four decimal places!**

Constant Dividend Growth: Yearly Increases

- During a year,

$$\frac{D}{1+r_q} + \frac{D}{(1+r_q)^2} + \frac{D}{(1+r_q)^3} + \frac{D}{(1+r_q)^4}$$

- The Dividend Discount Model says that

$$P_0 \overset{\text{should}}{=} \frac{(1+g)D_0}{r-g} = \frac{D_1}{r-g}$$

- Awesome interpretation!

$$\overbrace{\frac{D_1}{P_0} + g}^{\text{Specific Formula}} \overset{\text{should}}{=} r \overset{\text{should}}{=} \overbrace{\underbrace{\frac{D_1}{P_0}}_{\text{Didend Yield}} + \underbrace{\frac{P_1 - P_0}{P_0}}_{\text{Capital Gain Rate}}}^{\text{General Formula}}$$

- In this case, the <u>growth rate</u> g is the <u>capital gain rate</u>

Example: Stock Price

Microsoft (MSFT) requires a quarterly total return of 1.9402%. On 2017-11-01, it paid a dividend of 0.42 USD per share (Microsoft's dividends had been growning at an average quarterly rate of 1.87%). Based on this information, what should Microsoft's stock have traded for? (For reference, it actually traded for 84.17 USD.)

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$$P_0 \overset{\text{should}}{=} \frac{(1+g)D_0}{r-g} = \frac{0.42 + 1.87\% \times 0.42}{1.9402\% - 1.87\%} = 609.4786 \text{ USD}$$

Example: Implied Growth Rate

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$$g \overset{\text{should}}{=} \frac{rP_0 - D_0}{P_0 + D_0} = \frac{1.9402\% \times 84.17 - 0.42}{84.17 + 0.42} = 1.4341\%$$

Example: Dividend Yield

Microsoft (MSFT) requires a quarterly total return of 1.9402%. On 2017-11-01, it had an implied quarterly dividend growth rate of 1.4341%. Under the assumption of constant dividend growth, what was its implied dividend yield?

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$$\text{dividend yield} = r - g = 1.9402\% - 1.4341\% = 0.5061\%$$